

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

DATE:

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

The screenshot shows a web browser window with a Gmail 'Original Message' page. The browser tabs include 'mail login - Search', 'Is AI ruining your IT stock portfolio', 'Original Message', 'You are signed in as 99240041262', and 'Google'. The address bar shows the URL: <https://mail.google.com/mail/u/1/?ik=c0e6d2c505&view=om&permmsgid=msg-f:1874203670874351406>. The 'Original Message' section displays the following headers:

Message ID	<ytfadh17873370838360377@updates.share.market>
Created at:	Sat, Aug 22, 2026 at 11:55 AM (Delivered after 1 second)
From:	"Share.Market" <no-reply@updates.share.market> Using NetcoreCloud Mailer
To:	vivek426761@gmail.com
Subject:	Is AI ruining your IT stock portfolio?
SPF:	PASS with IP 175.158.69.185 Learn more
DKIM:	'PASS' with domain updates.share.market Learn more
DMARC:	'PASS' Learn more

Below the headers are two buttons: 'Download Original' and 'Copy to clipboard'. The bottom section shows the raw email source code, starting with 'Delivered-To: vivek426761@gmail.com' and including various technical details like 'Received: by 2002:a05:6200:5108:20b0:658:ccce:5fb9 with SMTP id t8-n2csp1984160qna; Fri, 21 Aug 2026 23:25:04 -0700 (PDT)'. The Windows taskbar at the bottom shows the date as 23-08-2026 and time as 22:16.

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.

- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - The sending server's hostname/IP address.
 - The receiving server's hostname/IP address.
 - The date and time the email passed through.

The screenshot shows the MXToolbox website's 'Email Header Analyzer' tool. The browser tabs include 'mail login - Search', 'Is AI ruining your IT stock portfolio', 'Original Message', 'You are signed in as 99240041262', and 'Email Header Analyzer, RFC822 Pa'. The URL is 'https://mxtoolbox.com/EmailHeaders.aspx?_gl=1*hz3iy0*_up*MQ..*_gs*MQ..&gclid=EALaQobChMI9Sispu3lgMvIw6DAx1wQSF0EAAAYASAAEgK_t_D_BwE&gbrad=0AAAAAD_qKm...'. The tool's navigation bar includes 'SuperTool', 'MX Lookup', 'Blacklists', 'DMARC', 'Diagnostics', 'Email Health', 'DNS Lookup', and 'Analyze Headers'. The 'Analyze Headers' section is active, showing a text area with a pasted email header. Below the text area is an 'Analyze Header' button. The header text includes: 'Delivered-To: vivek426761@gmail.com', 'Received: by 2002:a05:6200:5108:20b0:658:ccee:5fb9 with SMTP Id t8-n2csp1984160qna; Fri, 21 Aug 2026 23:25:04 -0700 (PDT)', 'X-Received: by 2002:a05:6a00:b88:b0:845:bda6:574b with SMTP Id d2e1a72fcca58-851f9aa2dfbmr16738175b3a.5.1787379904587; Fri, 21 Aug 2026 23:25:04 -0700 (PDT)', 'ARC-Seal: i=1; a=rsa-sha256; t=1787379904; cv=none; d=google.com; s=arc-20260327; b=DjilLyHgF9DBFEol89vC59mKo3VCPKaLIIIZva/Z2P33iBoK/5+ehCiwUlmSmyU6O47+gITbsvBdj+rkfBDH2zCJgcd7M77NadoMxaccKcCbakA3Js593MKcOC5xJRp++g Ov+OYY41uL0aRjU1GJ0hdK1BCcQAQA0omUSm35p0pX8+6MM+hg+HgnOCQUpMw5bw/ik+Nyi+m+KshboDBq9nxxGEu6J2VgqrhJdA6lrMqLy7BpHrmBWg8vjTtnVa9E/Umtqog yEkmtJAp1c046kMqpoyaRzYUC/ZdoRXwygZZ7IghRyo/7IPQhzbDBF2H9RaWTMQPNhK Rslg==', 'ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe-post:feedback-id:list-unsubscribe :content-transfer-encoding:mime-version:message-id:subject:from:to :date:dkim-signature:dkim-signature; bh=ClKyeTmK2hnZp10w/6BplepiwMpl4nvuc0rD8ZIY58U=; fh=ANBWwXvLhKep9zTOOTteR5OTlygvXOjBu7VHYDZ3U=; b=Ge3ZOifHdQ8nR9q0CMu/lkUdHcUF18wF04+2GIUYwNUhnrFBsGR5BE99FQs5F7pEHxz 6ye80iPwrE/zRsMTwScWjotEbYAJheu4OABpNYIh9XsEsgnGo2fxLFUN328NvMEaVhB X73bjGuWlaxVHZwitoMnpM8uYdYXQYFMz+fidAnkVanktXgrfUyaf+R3/DvxL66u44 I5kwUoy7T7dX2mOohGDBUldJYyFsfYnCCfyibwpSrNMAeaOPJM0a0P4xwjPFSILKfgh', and 'X-SPF: Pass'. Below the text area is an 'Analyze Header' button. The 'ABOUT EMAIL HEADERS' section explains that the tool makes email headers human readable by parsing them according to RFC 822. The footer shows 'Your IP is: 103.238.230.194 | Contact Terms & Conditions Security API Privacy Phone: (866)-696-6652 | © Copyright 2004-2026, MXToolbox, Inc. All rights reserved. US Patents 10839353 B2 & 11461738 B2'. The system tray shows '29°C Partly cloudy', a search bar, and the date '23-08-2026'.

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.

- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

The screenshot shows the MXToolbox Email Header Analyzer interface. The browser address bar displays the URL: `https://mxtoolbox.com/Public/Tools/EmailHeaders.aspx?huid=f7437de8-bd54-4793-8f23-70434db40c32`. The page title is "SPF and DKIM Information".

DMARC Section:

- Header: `dmARC:updates.share.market` with a "Show" button and a "Solve Email Delivery Problems" button.
- DMARC Record for `updates.share.market`:
 - Message: `No_DMARC_Record_found_for_sub-domain.`
 - Text: "Organization Domain of this sub-domain is: share.market. Inbox Receivers will apply share.market DMARC record to mail sent from updates.share.market"
 - Text: "SP Tag ' ' found: Inbox Receivers will treat all mail sent from updates.share.market that fails DMARC as suspicious."
- DMARC Record for `share.market` (organizational domain):
 - Record: `v=DMARC1;p=reject;rua=mailto:dmARC-alerts@share.market;pct=100`

SPF Section:

- Header: `spf:delivery.updates.share.market:175.158.69.185` with a "Show" button and a "Solve Email Delivery Problems" button.
- Record: `v=spf1 ip4:202.162.224.0/19 exists:%{ir}._spf.netcorecloud.net ~all`

DKIM Section:

- Header: `dkim:updates.share.market:nc2048` with a "Show" button.
- Record: `DKIM Public Record: [Redacted]`

The footer of the page includes: "Your IP is: 193.228.220.184 | Contact Terms & Conditions Security API Privacy Phone: (866)-696-6652 | © Copyright 2004-2026, MXToolBox, Inc. All rights reserved. US Patents 10839353 B2 & 11461738 B2". The Windows taskbar at the bottom shows the date as 23-08-2026 and time as 22:26.

7. Analyze the ‘Message-ID’

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.
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Example of a Header Analysis

Here's a quick example with a sample header:

Copy code

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

mail login - Search

Is AI ruining my IT stock portfolio?

Original Message

You are signed in as 99240041262

Email Header Analyzer, RFC822 P1

https://mxtoolbox.com/Public/Tools/Email/headers.aspx?huId=F7437de8-bd54-4793-8f23-70434db40c32

Chat

Header Name	Header Value
Delivered-To	vivek426761@gmail.com
X-Received	by 2002-a05-5a00-b88-b0 845-bda6-574b with SMTP id d2eta72lcca58-8519aa2d2bmr16738175b3a.5.1787379904587; Fri, 21 Aug 2006 23:25:04 -0700 (PDT)
ARC-Seal	i=1, a=sas-sha256, t=1787379904, cv=none, d=google.com, s=sarc-202603027, b=DjLhYg9DBFE0i89vC59mk0a3vCPKaLiRZvaLZP33iB0K5+ehCvUimsSmYU0 47-gI7BsvB4q+KBDH2zCjM7T7NadoMvacKcEbaKa3i593MKOC5XlRp+g Oy+OvY41uL0aRiUJ10hMK1BCCoQA0A0mU5m35p0 X8-EMM-hg-HgMOCQLpM5w5el k-NyIm-n-KshboDBqtrXcEu6J2VgqHJDA6mQy7BpHmBwgbY7mV9AEUjmTqog yEKmTJAp1c046MMapoyaRZYUCZd0XxwvZ27TgphR7i0P0hbcDBF2H9RAwTmQPNNK Rslg==
ARC-Message-Signature	i=1, a=sas-sha256, c=relaxed+relaxed, d=google.com, s=sarc-202603027, b-list-unsubscribe-post-feedback-icid-list-unsubscribe content-transfer-encoding mime-version message-id subject from to .date dkim-signature dkim-signature, bh=CkYekTmk2hZp10w9BplepMpl4muc0rD8Zy58U=: th=IANBWwXvLhK ep9zTOOT7R50TYpXQ0U7VhYDZG3=: b=Ge3z0H0Q8nR9q0CmuH0UcUf18wF04+2GlyWvNUNhBtsGR5E99FQz5F7EHz6: 6y80PwEzR3mTmSwjEYbaHeu4OABpNYm9XEs9nGoZ6LrUN328NwMEAvb X73bGuWlaxVhZwtoMhpM8uUdYLXQYFMz=fdAnkVankX0grUyat-R3DvL66u4 4l5kuU0y77FyYfYtCCYbwp5fNMAeaOPJMOa0F4wvJPF3ILKfgh jJGozqzPKZFdyH0uaSGLRA4L+uLReuAuSvpl9q6p848C4J1B+nS17nhcKJZPbuKCr BKGQ==: dariangoogle.com
ARC-Authentication-Results	i=1, mx.google.com, dkim-pass header i=@updates share market header s=nc2048 header b=clZyHnkD, dkim-pass header i=@env.etransmail.com header s=nc-fnc header b=RNsRtVM; spf-pass (google.com: domain of 17873730838360377-181468-1-gmail.com@delivery.updates.share.market designates 175 158 69 185 as permitted sender) smtp.mailfrom=17873730838360377-181468-1-gmail.com@delivery.updates.share.market; dmarc-pass (google.com: REJECT spf=REJECT dis=NONE) header=from=share.market
Return-Path	<17873730838360377-181468-1-gmail.com@delivery.updates.share.market>
Received-SPF	pass (google.com: domain of 17873730838360377-181468-1-gmail.com@delivery.updates.share.market designates 175 158 69 185 as permitted sender) client-ip=175 158 69 185;
Authentication-Results	mx.google.com, dkim-pass header i=@updates share market header s=nc2048 header b=clZyHnkD, dkim-pass header i=@env.etransmail.com header s=fnc header b=RNsRtVM; spf-pass (google.com: domain of 17873730838360377-181468-1-gmail.com@delivery.updates.share.market designates 175 158 69 185 as permitted sender) smtp.mailfrom=17873730838360377-181468-1-gmail.com@delivery.updates.share.market; dmarc-pass (google.com: REJECT spf=REJECT dis=NONE) header=from=share.market
DKIM-Signature	v=1, a=sas-sha256, c=relaxed+relaxed, d=updates.share.market, s=nc2048, b-list-unsubscribe-post-feedback-icid-list-unsubscribe content-transfer-encoding mime-version message-id subject from to from to subject, bh=CkYekTmk2hZp10w9BplepMpl4muc0rD8Zy58U=: b=clZyHnkDhLXds59 Bp8z8L3d465cN2RvCYKf1H55Ca8V9rShmm1G03pU0c708Y7YLYLwA+weCW1u0HbX2YKf6M6-LW7Z5Y5Jsvm3gVntfXmZqemB9H1/Qwv176LmYm1mpm+VGNd8H23D4BGmCA.wY39pzeK0K0TRbQHT3B54QbE1FYB1YdXQLgZvTmcs2PmHGZX+1F04d+Ch0cDT5214b4A1 1XPW7Y7X8ZHO yC58VYL3P30y0IobXk0AThB6z0rhm+806gU44v9z8hm1jJ0rD0rInObR8vYRY dZ0Rdy0AAwQvJkpp5310dHvUdu==
Date	Sat, 22 Aug 2006 11:55:03 +0530
To	<vivek426761@gmail.com>
From	Share Market <no-reply@updates.share.market>
Subject	Is AI ruining your IT stock portfolio?
Message-ID	<yftadh17873730838360377@updates.share.market>
MIME-Version	1.0
Content-Type	multipart/alternative, boundary="-----1787379830187815665"
Content-Transfer-Encoding	binary
List-Unsubscribe	<http://delivery.updates.share.market/REUIVLGAkZ?id=181468-N0sBAIRUJgEMHREVERISRJBfXFRREEYUuQ0BBGUVCfXETRRRGfRBERURENJEEfLQgwSBFMFvVpYEBB5AgWFF9LWwIsFAGDQYBAVMcVQGMVURCwZ3TfBEFukWwBgeVvBUdF0EGreJRWBYURberW0DABKVEId WUKBFXNZXrmt.G4qChUgavPYV4SAQ==>, <mailto:17873730838360377-181468-fc0unsb@usub.iframe03.com?subject=Unsubscribe>
Feedback-Id	181468 202608 1 falconide
X-Dkim-Options	
X-Fncid	181468-17873730838360377-0
X-Intline	1787379830
X-Traffic-Type	181468-2
X-Mailer	NetcoreCloud Mailer
List-Unsubscribe-Post	List-Unsubscribe=One-Click

29°C
Partly cloudy

22:27
23-08-2026

ENG IN

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<img src=3D"https://delivery.updates.share.market/REUIVL0AKZin=3D30181468=3D=LisB41ag7tMhVEt8R78FwXnREUYQQ8G8GUVc=3DTranGf=3DREURRNN1E11LQg=3DfW=3DfWvP8154g9hWl9Lw1aSFAGQdYBAPVHCQ9V12RcuZ5T7BFFUkH8g6vWbDfREg80RE78BU4YER80ABKVeTdHUIKXFNZXRmLGA4cHsqwpYVK4SAQ=3D=3D" alt=3D"" /></body>
</html>
--_-----e_1787379830187815665--
```